

Sri Sai Mamidi | Data Engineer

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SUMMARY

Data Engineer with **5+ years of experience** building scalable data pipelines, cloud data warehouses, and real-time ETL frameworks across insurance, healthcare, and enterprise domains. Proficient in AWS, Azure, Snowflake, Airflow, Kafka, and PySpark with expertise in data modeling, CDC, and streaming solutions. Skilled in implementing validation frameworks, CI/CD automation, and governance practices ensuring secure, compliant, and reliable data delivery. Known for reducing latency, improving efficiency, and enabling actionable analytics supporting risk modeling, compliance reporting, and operational decision-making.

PROFESSIONAL EXPERIENCE

Liberty Mutual Insurance, TX, US

Mar 2024 - Current

Data Engineer

- Developed AWS Glue and PySpark pipelines processing 20M+ weekly insurance records into Snowflake, reducing data delivery latency 65% and improving risk modeling accuracy, underwriting analytics efficiency, and reporting timelines.
- Designed star schema warehouse models, reducing Tableau query execution by 40% and enabling actuarial teams to analyze loss trends faster, optimize pricing strategies, and support informed premium adjustment decisions.
- Automated Apache Airflow DAGs with SLA monitoring, retry handling, and dependency management, increasing reliability 35%, minimizing overnight failures, and ensuring uninterrupted availability for downstream reporting and analytics across actuarial teams.
- Created Python-based validation scripts enforcing schema integrity, null handling, and anomaly checks, reducing QA escalations 30% while enhancing trust in analytics outputs and strengthening enterprise-level data governance practices.
- Integrated AWS Secrets Manager and IAM roles with security teams, achieving 100% audit compliance, safeguarding sensitive data, and ensuring fine-grained access control across Liberty Mutual's enterprise data ecosystem.

CitiusTech, India

Jul 2018 – Aug 2021

Data Engineer

- Built ingestion pipelines using Azure Data Factory, Kafka, and PySpark to integrate HL7 and FHIR healthcare records into Snowflake and Synapse, improving analytics readiness 40% across diverse enterprise environments.
- Standardized reusable transformation workflows using dbt, enforcing consistent KPI definitions and dimensional modeling, eliminating metric duplication, and enabling accurate clinical, compliance, and financial reporting across 10+ healthcare enterprise clients.
- Migrated 25+ SSIS packages into Airflow DAGs with scheduling, retries, and monitoring, reducing runtime 4 hours daily, improving refresh timeliness, and enhancing healthcare data availability across operational pipelines.
- Optimized incremental loads and CDC processes using timestamp-based watermarks and record versioning, enabling seamless processing of 50M+ monthly healthcare records, reducing duplication, and ensuring timely data delivery for analytics.
- Developed Python validation scripts to handle schema mismatches, null fields, and anomalies, reducing ingestion failures 25%, strengthening compliance, and ensuring downstream reporting accuracy for enterprise healthcare data pipelines.
- Established GitHub-based version control and automated CI/CD pipelines with GitLab and GitHub Actions, streamlining deployments, enabling rollback triggers, and improving environment consistency across development, QA, and production.
- Partnered with BI, DevOps, and compliance teams to define SLAs, data contracts, and onboarding documentation, standardizing delivery expectations, reducing escalations, and accelerating enterprise healthcare client integration timelines.

Intex Technologies, India

Oct 2017 – Jun 2018

Jr. Data Engineer

- Automated SQL pipelines loading ERP-based sales and inventory data into Redshift, reducing manual reporting effort 40% and enabling near real-time dashboards supporting supply chain and operational analytics.
- Scheduled Bash-driven ETL jobs with cron, maintaining 95%+ availability for reporting datasets, ensuring timely insights, and supporting operational teams with reliable sales and distribution performance metrics.
- Enhanced Python integration scripts to normalize REST API JSON responses, increasing data coverage 60%, enabling richer business dashboards, and improving leadership visibility into sales and product distribution performance.
- Authored schema documentation, transformation mappings, and process flow diagrams, reducing analyst onboarding time 50% while improving transparency and cross-functional collaboration between engineering, BI developers, and reporting stakeholders.
- Collaborated with BI teams to validate transformation outputs, resolve data logic issues, and accelerate report delivery cycles, improving on-time reporting accuracy by 25% across sales and operations teams.

EDUCATION

Masters in Information Technology | Trine University, Angola, Indiana

Aug 2021 – Dec 2023

Bachelors in Business Administration | Osmania university, Hyderabad, India

Jun 2016 – May 2019

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Bash

Cloud Platforms: AWS (S3, Redshift, Glue, Lambda), Azure (Data Factory, Synapse), GCP (BigQuery)

ETL & Data Pipelines: Apache Airflow, AWS Glue, dbt (basic), SSIS

Data Warehousing & Storage: Snowflake, Amazon Redshift, Google BigQuery, PostgreSQL, MySQL, MongoDB

Big Data Tools: Apache Spark (PySpark), Hive, Hadoop (basic understanding)

Data Modeling: Star Schema, Snowflake Schema, Dimensional Modeling, ER Diagrams

Version Control & DevOps: Git, GitHub, GitLab, Jenkins (basic), GitHub Actions

Workflow Orchestration: Apache Airflow, Cron Jobs

APIs & Integration: REST APIs, JSON, XML, Postman

Data Quality & Governance: Great Expectations (basic), data validation scripts

Security & Compliance: IAM (AWS/Azure), Encryption (at rest/in transit), basic HIPAA/GDPR awareness

Monitoring & Logging: AWS CloudWatch (basic), Datadog (introductory)

Soft Skills & Agile: Agile/Scrum, Team Collaboration, Communication